Philips
Healthcare

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data speaks

Predictive Maintenance- POC

Case Study

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Predictive Maintenance for Medical Equipment Failures and Alert Management

Problem Statement

Objective:

- Build a predictive maintenance system that empowers dashboards with medical equipment failure predictions and fault alert management, enabling proactive biomedical maintenance planning.

Challenges addressed:

- Unplanned downtime due to sudden medical device failures
- Lack of early visibility into equipment fault triggers and calibration drift
- Inconsistent telemetry data quality across device models and hospital sites
- Difficulty in scaling predictive models across diverse equipment fleets and modalities

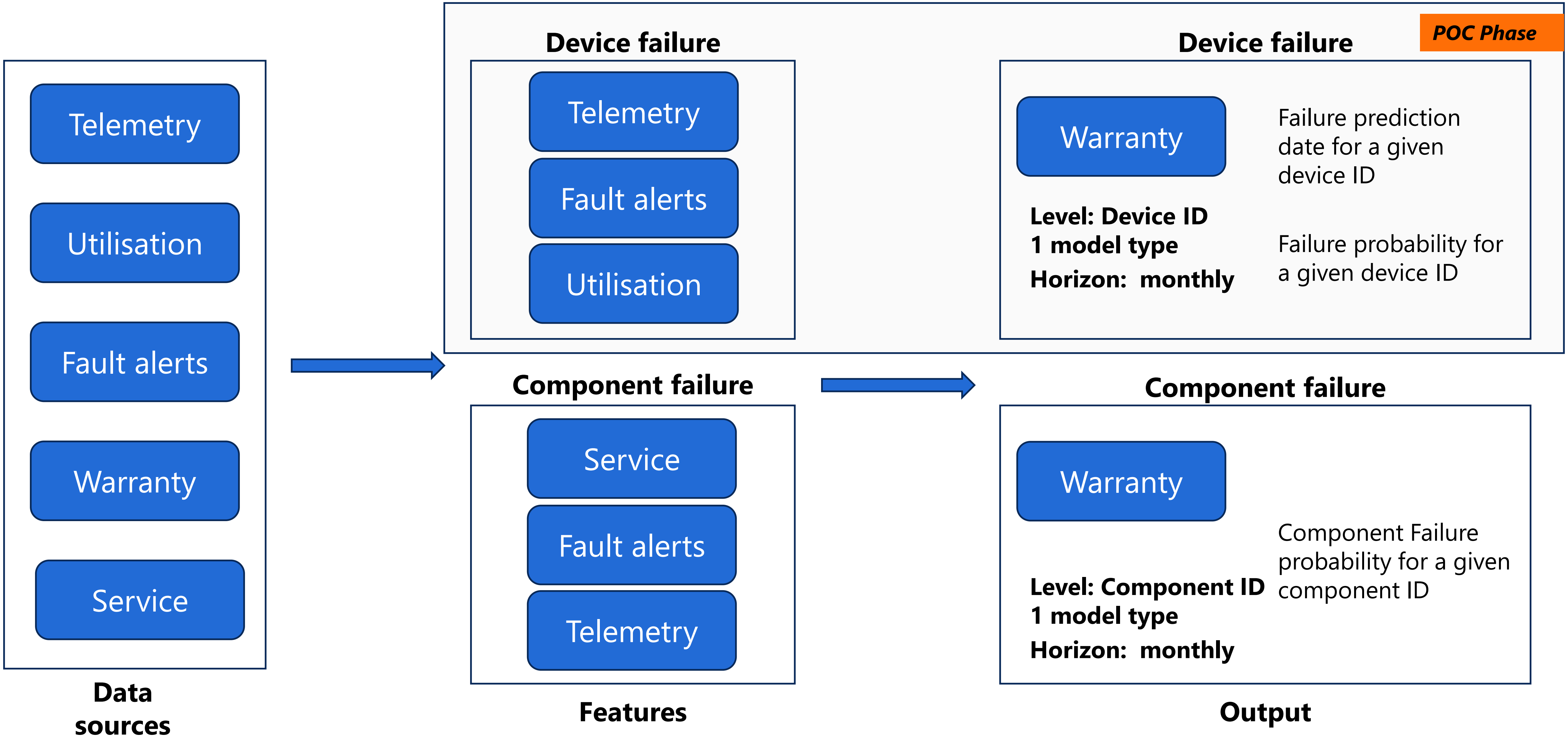
Solution

- Data preparation and cleaning of device telemetry, usage logs, fault codes, and service records
- Exploratory data analysis to identify device failure patterns and critical signal features
- Iterative QA with validation metrics (WAPE, MAPE, FA) and back-testing (predicted vs actual failures)
- Deployment of the best-fit model integrated into maintenance dashboards for real-time predictive alerts

Outcome and Impact

- **Outcome:**
  - Early detection of potential device failures and fault code occurrences
  - Reduced equipment downtime and improved hospital operational continuity
- **Impact:**
  - Enhanced biomedical decision-making through predictive dashboards
  - Scalable framework that can adapt to weekly and monthly forecast horizons

Proposed Predictive Maintenance for POC - illustration



Predictive Maintenance process is devised while constantly assessing results, measuring impact, making clinical changes and providing feedback loop for improvement

Data preparation module

1 Hypothesis and data preparation
- Hypothesis generation
- Data gathering
- Data cleaning
- ETL setup

2 Exploratory Data Analysis (EDA)
- Understanding device performance
- Variable selection and Feature engineering

3 Analytical dataset creation
- Standardization
- Variable transformations
- Feature selection

Predictive Maintenance Solution

4 Model building process

Algorithm Development

- Identify Key Features
- Feature Optimization
- Parameter Selection
- Model Training
- Deployment & Integration

(iii) Model development

ML models
1. Multinomial Logistic Regression
2. Random Forest
3. CatBoost
4. LightGBM
5. XGBoost

Dimensionality Reduction
1. Multiple Correspondence Analysis – Categorical Variables
2. PCA – Numerical Variables

Different models will be developed, and their accuracies and biases will be checked, post which the best model will be selected considering specific approach for each model;

Iterative

QA module

5 Validation & Iterations
- WAPE/FA/ MAPE calculations

6 Predictions
- Prediction Date
- Probability of maintenance

7 Measurement
- Back testing
- Prediction vs Actual

Please note: *this is the generic procedure followed for Predictive Maintenance, and selective steps will be followed as applicable*

The process for building weekly and monthly models remains the same while nuances, based on forecast horizon, will be captured by the algorithms

Level of predictive maintenance and the features considered for Device failure

Medical device x model (5 types) x device id	MRI Scanner
	CT Scanner
	Ultrasound System
	Patient Monitor
	Ventilator

Features

Performance	Operating Hours (Period Hrs) from utilisation data	numeric
	Scan/Procedure Time (Period in hrs)	numeric
	Idle Time Band (Period in hrs)	numeric
	Active Use Band (Period in hrs)	numeric
	Peak Load Band (Period in hrs)	numeric
	Imaging Job (Hrs.)	numeric
	Calibration Job (Hrs.)	numeric
	Maintenance Job (Hrs.)	numeric
	Emergency Job (Hrs.)	numeric

Utilization	Device Model	Categoric
	Hospital / Site	Categoric
	Modality	Categoric
	Runtime Hours (Period Hrs)	numeric

Fault Alerts	Device_Runtime_Hours	numeric
	FaultAlertGenerationTime	convert to numeric
	FaultAlertClosureTime	convert to numeric
	FaultErrorCode	Categoric

Service	Count of PM services before failure date	numeric